University of Sharjah
College of Sciences
Department of Applied Biology
Biotechnology Program
Four Years Plan of study
AY 2026/2027

جامعة الشارقة
UNIVERSITY OF SHARJAH

University of Sharjah / College of Sciences / Department of Applied Biology / Biotechnology Program / Four Years Plan of Study
Note: This study plan is applicable to students who joined the program in Fall 2025 or later.

Fall/First Year				Spring/First Year			
Course No.	**Course Name**	**Cr. H**	**Prerequisite**	**Course No.**	**Course Name**	**Cr. H**	**Prerequisite**
0104100	Islamic Culture	3	-	1501110 (E) 1501111 (A)	AI and Digital Technologies	3	-
0201102 0201105	Arabic Language or Arabic Language for non-Arabic Speakers	3	-	1430107	General Physics for (HS) *	4	-
0202112	English for Academic purpose	3	-	1440163	Calculus for (HS)	3	-
1426155	General Chemistry (HS) *	4	-	1450102	General Biology (2) *	3	1450101
1450101	General Biology (1) *	3	-	XXXXXX	University Elective (1) One course from **Domain F**	3	-
1450107	General Biology Lab *	1	1450101 or Cor				
	Total	**17**			**Total**	**16**	
Fall/Second Year				**Spring/ Second Year**			
Course No.	**Course Name**	**Cr. H**	**Prerequisite**	**Course No.**	**Course Name**	**Cr. H**	**Prerequisite**
1511116	Programming	4	-	0202121	English for Medical students	3	0202112
0504252	Biostatistics	3	-	1420225	Analytical Chemistry	3	1426155
1426217	Organic Chemistry (HS) *	4	1426155	1420226	Analytical Chemistry Lab	1	1420225 or Cor
1450202	Basic Biotechnology	3	1450102	1450251	Cell Biology *	3	1450102
0204102	UAE Society	3	-	XXXXXXX	University Elective (2) One course from **Domain G**	3	-
				XXXXXXX	University Elective (3) One course from **Domain H**	3	-
	Total	**17**			**Total**	**16**	
Fall/Third Year				**Spring/ Third Year**			
Course No.	**Course Name**	**Cr. H**	**Prerequisite**	**Course No.**	**Course Name**	**Cr. H**	**Prerequisite**
1450331	General Microbiology	3	1450251	1450302	Bioinformatics	3	1450202
1450332	General Micro. Lab	1	1450331 or Cor	1450303	Bio-informatics Lab	1	1450302
1450333	Immunology & Serology	3	1450251	1450351	General Biochemistry *	3	1426217+1450251
1450336	Immunology and Serology Lab	1	1450333 or Cor	1450352	General Biochemistry Lab *	1	1450351 or Cor
1450341	Molecular Genetics	3	1450251	1450361	Tissue Culture & Animal Biotech.	2	1450332+1450333
1450342	Molecular Genetics Lab	1	1450341 or Cor	1450XXX	Departmental Elective (2)	3	
1450XXX	Departmental Elective (1)	3		1450XXX	Departmental Elective (3)	3	
	Total	**15**			**Total**	**16**	

1450399: Field training: Students will spend 6 weeks during the summer/ 3 Cr. H. Prerequisite: 65 Cr. H +1450361+1450352

A-Forensic & Medical Biotechnology Concentration

Fall/ Fourth Year				Spring / Fourth Year			
Course No.	**Course Name**	**Cr. H**	**Prerequisite**	**Course No.**	**Course Name**	**Cr. H**	**Prerequisite**
1420429	Forensic Chemistry	3	**1420225**	1450311	Introduction to Forensic Sciences	3	1450102+1426155
1450451	Molecular Biology	3	1450341	1450441	Molecular Human Genetics	3	1450341
1450452	Molecular Biology Lab	1	1450451 or Cor.	1450491	Seminar	1	Completion 90 C.H.
1450455	Biotechnology in Medicine	3	1450341	1450493	Research Project	3	Completion 99 C.H.
1450XXX	Departmental Elective (4)	3		1450496	Ethics in Biotechnology	1	1450202
	Total	**13**			**Total**	**11**	

B-Industrial Biotechnology Concentration

Fall/ Fourth Year				Spring / Fourth Year			
Course No.	**Course Name**	**Cr. H**	**Prerequisite**	**Course No.**	**Course Name**	**Cr. H**	**Prerequisite**
1450431	Microbial Processing & Biotechnology	4	1450202+1450331	1450461	Animal Biotechnology	2	1450361
1450451	Molecular Biology	3	1450341	1450471	Industrial Biotechnology	3	1450331
1450452	Molecular Biology Lab	1	1450451 or Cor.	1450491	Seminar	1	Completion 90 C.H.
1450453	Protein Biochemistry & Engineering	3	1450351	1450493	Research Project	3	Completion 99 C.H.
1450XXX	Departmental Elective (4)	3		1450496	Ethics in Biotechnology	1	1450202
	Total	**14**			**Total**	**10**	

C-Plant & Environmental Biotechnology Concentration							
Fall/ Fourth Year				Spring / Fourth Year			
Course No.	Course Name	Cr. H	Prerequisite	Course No.	Course Name	Cr. H	Prerequisite
1450421	Plant Biotechnology	4	1450202	1450433	Marine Biotechnology	3	1450202+1450331
1450436	Environmental Biotechnology	3	1450202+ 1450331	1450442	Genetically Modified Organisms	2	1450202
1450451	Molecular Biology	3	1450341	1450491	Seminar	1	Completion 90 C.H.
1450452	Molecular Biology Lab	1	1450451 or Cor.	1450493	Research Project	3	Completion 99 C.H.
1450XXX	Departmental Elective (4)	3		1450496	Ethics in Biotechnology	1	1450202
	Total	**14**			**Total**	**10**	

University of Sharjah
College of Sciences
Department of Applied Biology
Bachelor Biotechnology Program

University of Sharjah / College of Sciences / Department of Applied Biology / Biotechnology Program /BSB Compulsory Courses
(For all concentrations)

University Required (Compulsory) Courses (15 C.H)

Course No.	Course Name	C.H	Prerequisite
0104100	Islamic Culture	3	-
0201102 0201105	Arabic Language or Arabic Language for non-Arabic Speakers	3	-
0202112	English for Academic Purpose	3	-
0204102	UAE Society	3	-
1501110 (E) 1501111 (A)	AI and Digital Technologies	3	-
	Total	15	-

College of Sciences Required (Compulsory) Courses (15 C.H)

Course No.	Course Name	C.H	Prerequisite
1511116	Programming 1	4	-
1426155	General Chemistry for Health Sciences *	4	-
1430107	General Physics for Health Sciences *	4	-
1440163	Calculus 1 for Health Sciences	3	-
	Total	15	

Supporting Required (Compulsory) Courses offered by other departments (14 C.H)

Course No.	Course Name	C.H	Prerequisite
0202121	English for Medical Sciences	3	0202112
0504252	Biostatistics	3	-
1426217	Organic Chemistry (HS) *	4	1426155
1420225	Analytics Chemistry	3	1426155
1420226	Analytical Chemistry Lab	1	1420225 or Cor
	Total	14	

Program Compulsory Courses (For all concentrations) (47 C.H)

Course No.	Course Title	C.H	Prerequisites
1450101	General Biology (1) *	3	-
1450107	General Biology Lab *	1	1450101 or Cor
1450102	General Biology (2) *	3	1450101
1450202	Basic Biotechnology	3	1450102
1450251	Cell Biology *	3	1450102
1450331	General Microbiology	3	1450331
1450332	General Micro. Lab	1	1450332
1450333	Immunology & Serology	3	1450333
1450336	Immunology and Serology Lab	1	1450336
1450341	Molecular Genetics	3	1450341
1450342	Molecular Genetics Lab	1	1450342
1450302	Bioinformatics	3	1450202
1450303	Bio-informatics Lab	1	1450302
1450351	General Biochemistry *	3	1426217+1450251
1450352	General Biochemistry Lab *	1	1450351 or Cor
1450361	Tissue Culture & Animal Biotech.	2	1450332+1450333
1450399	Field training	3	65 Cr. H +1450361+1450352
1450451	Molecular Biology	3	1450341
1450452	Molecular Biology Lab	1	1450451 or Cor.
1450491	Seminar	1	Completion 90 C.H.
1450493	Research Project	3	Completion 99 C.H.
1450496	Ethics in Biotechnology	1	1450202
	Total	**47**	

Program Compulsory Courses (For each concentration) (12 C.H) Fourth year

A-Forensic & Medical Biotechnology Concentration			
Course No.	Course Name	Cr. H	Prerequisite
1420429	Forensic Chemistry	3	1420225
1450311	Introduction to Forensic Sciences	3	1450102+1426155
1450441	Molecular Human Genetics	3	1450341
1450455	Biotechnology in Medicine	3	1450341
B-Industrial Biotechnology Concentration			
Course No.	Course Name	Cr. H	Prerequisite
1450431	Microbial Processing & Biotechnology	4	1450202+1450331
1450461	Animal Biotechnology	2	1450361
1450471	Industrial Biotechnology	3	1450331
1450453	Protein Biochemistry & Engineering	3	1450351
C-Plant & Environmental Biotechnology Concentration			
Course No.	Course Name	Cr. H	Prerequisite
1450421	Plant Biotechnology	4	1450202
1450436	Environmental Biotechnology	3	1450202+ 1450331
1450433	Marine Biotechnology	3	1450202+1450331
1450442	Genetically Modified Organisms	2	1450202

University Required (Elective) Courses (9 C.H.)

1 course from domain F, 1 course from domain G, and 3rd course from domain H

Course No	Course Title	C. H	Prerequisites
Domain F/(Humanities, Social & Arts)			
0103103	Islamic System	3	None
0104130	Analytical Biography of the Prophet	3	None
0201140	Introduction to Arabic Literature	3	None
0202130	French Language	3	None
0203100	Islamic Civilization	3	None
0203102	History of the Arabian Gulf	3	None
0203200	History of the Sciences among Muslims	3	None
0204103	Principles of Sign Language	3	None
0206102	Fundamentals of Islamic Education	3	None
0206103	Introduction to Psychology ***	3	None
0302150	Introduction to Business Administration	3	None
0308150	Introduction to Economics	3	None
0602246	Human Rights in Islam and International Declarations	3	None
*** **Required course for students enrolled in the Pre-Med Track to support MCAT preparation.**			
Domain G (Sciences and Technology)			
0401142	Man and the Environment	3	None
0406102	Introduction to Sustainability	3	None
0503101	Health and Safety	3	None
0505100	Understanding Disabilities	3	None
0505101	Fitness and Wellness	3	None
0507101	Health Awareness and Nutrition	3	None
1430101	Astronomy and Space Sciences	3	None
1450100	Biology and Society	3	None
Domain H (Life Skills and Personal Development)			
0103104	Professional Ethics in Islamic Sharia	3	None
0302200	Fundamentals of Innovation and Entrepreneurship	3	None
0308131	Personal Finance	3	None
0601109	Legal Culture	3	None
0800108	Media Literacy and Digital Society	3	None
0206100	Smart and Effective Learning Skills	3	None

Note:

- Every student should take all courses in domains A to E and one course from each of domains F, G and H.
- Students who select to study any program taught in Arabic cannot register for elective courses taught in English.

Program Elective Courses (12 C.H)

Course No.	Course Title	C.H	Prerequisites
1450261	Animal Biology	3	1450102
1450221	Plant Biology	3	1450102
1450334	Microbial Genetics	3	1450332+1450341
1450391	Scientific Writing & Presentation	1	0202103+1450202
1450437	Pharmaceutical Biotechnology	3	1450202
1450454	Molecular Developmental Biology	3	1450451
1450497	Selected topics in Biotechnology A	3	65 Cr. H
1450498	Selected topics in Biotechnology B	2	65 Cr. H
1450499	Selected topics in Biotechnology C	1	65 Cr. H
1450422	Biotechnology for Resources Sustainability	3	1450202+1450331
1450472	Food Biotechnology & Analysis	4	1450352
0501370	Hematology	4	1450251
1420103	General Chemistry II *	3	G Chemistry HS (1426155)
1420104	General Chemistry II Lab *	1	G Chemistry II (1420103) or Cor
1420217	Organic Chemistry II *	3	Org Chemistry HS (1426217)
1420218	Organic Chemistry II Lab *	1	Org Chemistry II (1420217) or Cor
1430117	Physics II *	3	G Physics HS (1430107)
1430118	Physics II Lab *	1	Physics II or Cor
1450252	Cell Biology Lab **	1	Cell Biology (1450251) or Cor

* Students who choose to pursue the Pre-Medical Track are required to complete the following designated courses. Together with the core Biotechnology curriculum, these courses satisfy common medical school prerequisite requirements and provide the academic preparation needed for the MCAT or equivalent medical school admission examinations.

**Non-Biotechnology students who choose to pursue the Pre-Medical Track are required to complete all courses marked with (*) in addition to the Cell Biology Lab (**), to satisfy common medical school prerequisite requirements and provide the academic preparation needed for the MCAT or equivalent medical school admission examinations.

Note: The Pre-Medical Track is designed for students planning to apply to medical schools that consider the MCAT or equivalent admission examinations, including international and UAE-based medical schools where applicable. Students are encouraged to check the specific admission requirements of their intended medical school, as requirements may vary